Long-Context Factual Reliability in Llama 3.2 3B Instruct

Experiment C: Total Inaccuracy Report

Model: meta-llama/Llama-3.2-3B-Instruct

Dataset: 100 question families, 500 repeated observations

Report date: 2026-08-10

Table of Contents

1. Executive Summary / Abstract

This report summarizes Experiment C, a completed local inference experiment evaluating factual reliability for meta-llama/Llama-3.2-3B-Instruct. The benchmark uses primary-source records from SEC filings, Drugs@FDA, ClinicalTrials.gov, and FRED to create 100 repeated question families evaluated at 4K, 8K, 16K, 32K, and 64K context lengths, for 500 total observations.

The main result is a strong increase in total inaccuracy as context length increases. Inaccuracy rose from 43% at 4K to 73% at 64K. In a GEE logistic model clustered by question family, each doubling of context length was associated with 1.35x the odds of an inaccurate answer (95% CI [1.20, 1.52], p = 4.06e-07). Mean inference latency increased from 0.317 s to 7.824 s.

2. Research Question and Motivation

The primary research question is how increasing context length affects factual reliability when a model is given legitimate but competing same-domain records. The report uses total inaccuracy as the main reliability metric: a response is either Correct or Inaccurate.

3. Dataset and Source Domains

The benchmark converts authoritative primary-source records into controlled factual questions. Each question family has a deterministic gold answer, target evidence, structured target conditions, and same-domain distractors. Contexts grow by adding plausible competing information while preserving the same question, gold answer, and target evidence.

Table 1. Source domains.

Domain	Role
SEC	Company financial facts and filings.
FDA / Drugs@FDA	Drug applications, submissions, products, strengths, dosage forms, and routes.
ClinicalTrials.gov	Trial metadata, dates, enrollment, arms, status, and unavailable fields.
FRED	Economic time series with unit, frequency, and series attributes.

4. Question Dataset

The final dataset contains 100 unique question families: 80 answerable and 20 unanswerable. Each family was evaluated at five context lengths, yielding 100 x 5 = 500 repeated observations. Because the same families appear at all context lengths, statistical tests treat observations as clustered by family.

Table 2. Question-type distribution.

Question type	Families	Purpose

DIRECT_RETRIEVAL	20	Single-record factual retrieval.
RETRIEVAL_CALCULATION	30	Retrieve values and compute a deterministic answer.
TEMPORAL_VERSION	11	Identify the requested period, version, or submission.
ENTITY_UNIT_BINDING	19	Bind entity, unit, product, route, dosage form, or series attributes.
UNANSWERABLE	20	Return the insufficient-evidence answer when target evidence is absent.

5. Experimental Configuration

Table 3. Inference configuration.

Setting	Value
Model	meta-llama/Llama-3.2-3B-Instruct
Model revision	0cb88a4f764b7a12671c53f0838cd831a0843b95
Hardware	NVIDIA GeForce RTX 4090
Precision/cache	BF16, standard DynamicCache
Decoding	deterministic greedy; do_sample=False; num_beams=1; max_new_tokens=128
Quantization/offloading	none
Prompt version/hash	llama_chat_v4 / 5d2869822989e19b
Output contract	ANSWER: <answer>

Experiment C achieved 500/500 successful inference instances, zero format failures, and zero outputs hitting the 128-token cap.

6. Grading Method

Responses were graded by a frozen deterministic grader (SHA-256 d9282a0ccc50daba3bfd232c058dfbab63a5c19a7323d585a7c2236b3a6c4ba8). Of 500 responses, 499 were resolved by the deterministic grader and one was resolved by documented human manual adjudication. The report presents total inaccuracy as the reliability outcome: an answer is Correct if it matches the deterministic gold answer after normalization, otherwise it is Inaccurate.

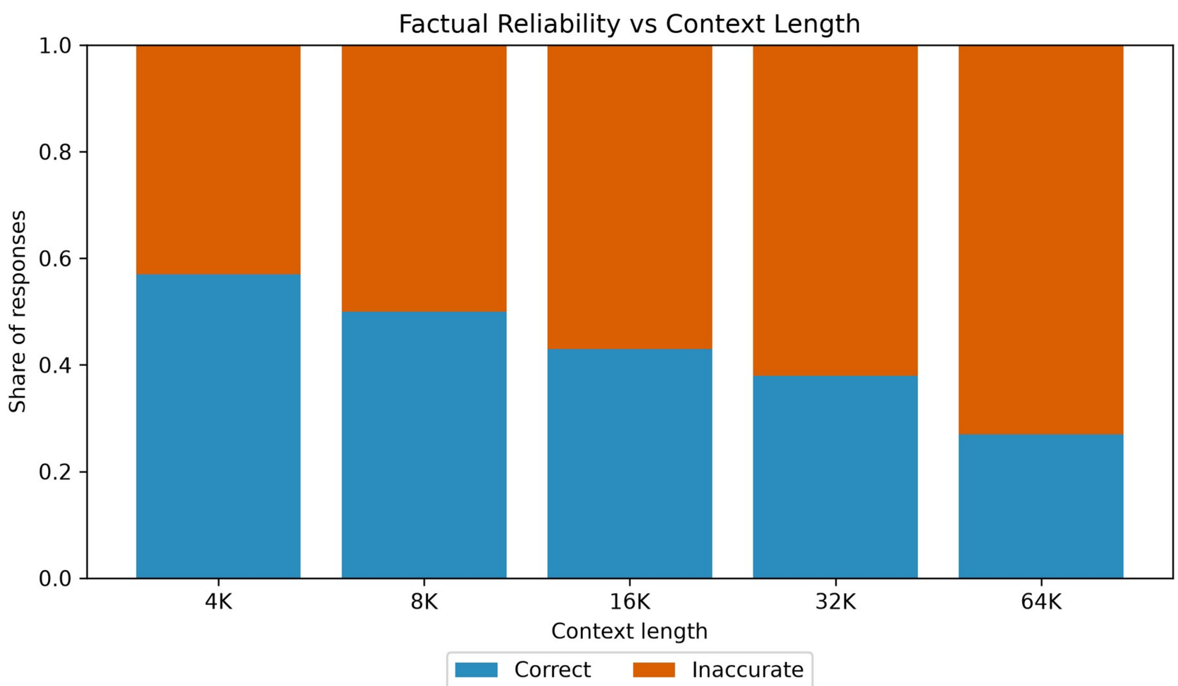
7. Overall Results

Table 4. Central factual-reliability results.

Context	N	Correct	Inaccurate	Inaccuracy 95% CI	Mean inference latency

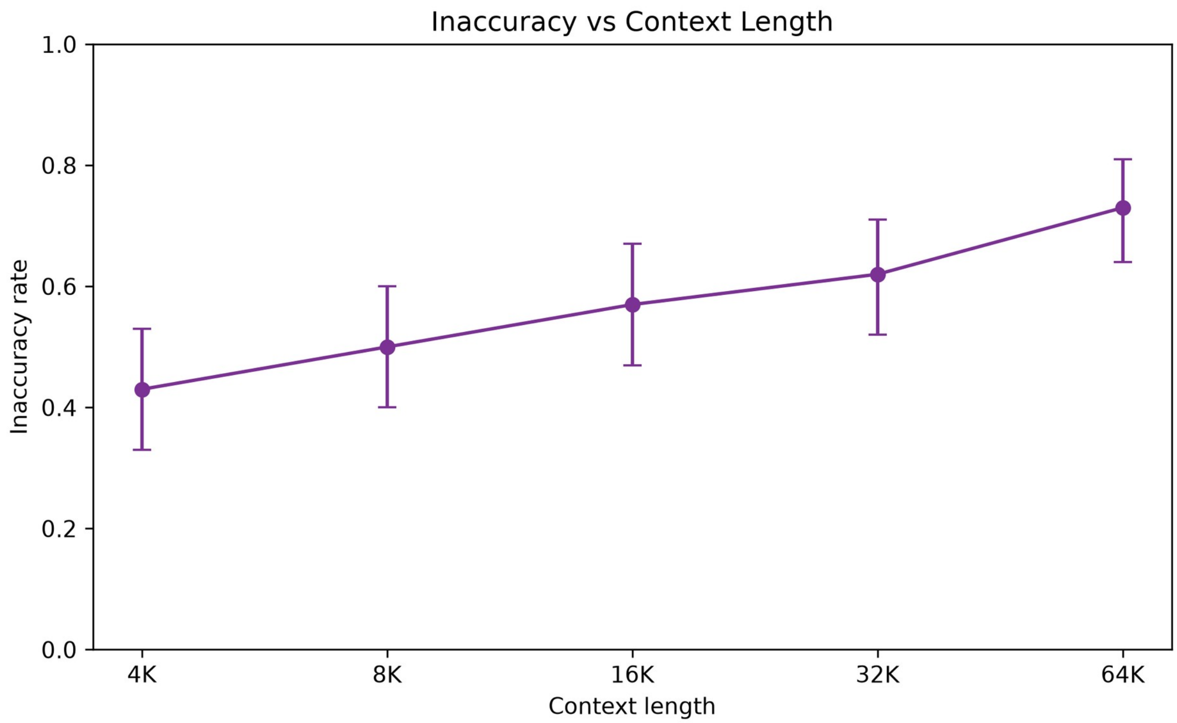
4K	100	57%	43%	[33%, 53%]	0.317 s
8K	100	50%	50%	[40%, 60%]	0.601 s
16K	100	43%	57%	[47%, 67%]	1.212 s
32K	100	38%	62%	[52%, 71%]	2.842 s
64K	100	27%	73%	[64%, 81%]	7.824 s

Figure 1. Factual reliability vs context length.



Correct and inaccurate responses sum to 100% at each context length.

Figure 2. Inaccuracy vs context length.



GEE OR per context doubling = 1.352, 95% CI [1.203, 1.519], p = 4.06e-07.

8. Statistical Analysis of Overall Inaccuracy

The repeated-measures model was fit as a GEE logistic regression clustered by question_family_id with context_log2 coded as 4K = 0, 8K = 1, 16K = 2, 32K = 3, and 64K = 4.

Accuracy OR per context doubling was 0.740 (95% CI [0.659, 0.831], p = 4.06e-07). Equivalently, inaccuracy OR per context doubling was 1.352 (95% CI [1.203, 1.519], p = 4.06e-07). This is an odds increase, not a percentage-point increase.

9. Paired Context Comparisons

Paired McNemar tests compared each longer context to 4K within the same question families. Holm correction was applied across the four comparisons.

Table 5. Paired inaccuracy comparisons.

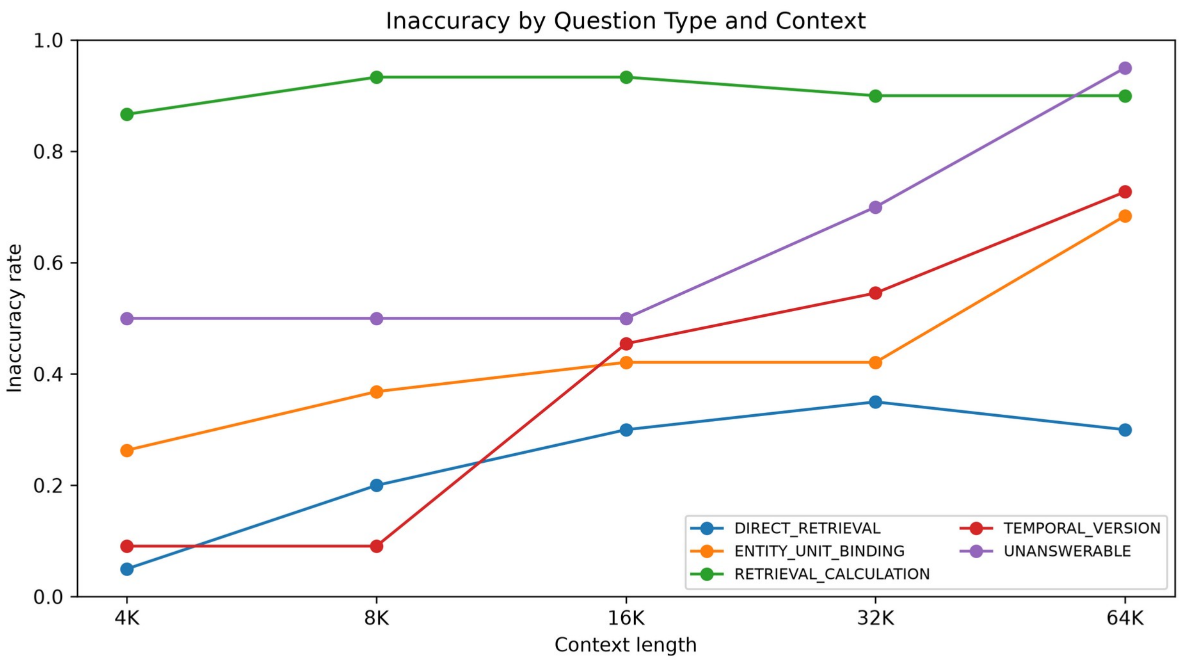
Comparison	Inaccuracy difference	Discordant pairs	Raw p-value	Holm-adjusted p-value	Result
4K vs 8K	7.0%	3/10	0.0923	0.0923	not significant after Holm
4K vs 16K	14.0%	3/17	0.0026	0.0052	p < 0.05 after Holm
4K vs 32K	19.0%	5/24	5.46e-04	0.0016	p < 0.05 after Holm

4K vs 64K	30.0%	5/35	1.38e-06	5.53e-06	p < 0.05 after Holm

10. Question-Type Results

Question-type analyses are exploratory because subgroup cell sizes are smaller than the primary repeated-measures analysis.

Inaccuracy by Question Type & Context.



Subgroup results are exploratory because cells are smaller than the primary repeated-measures analysis.

Table 6. Inaccuracy by question type and context.

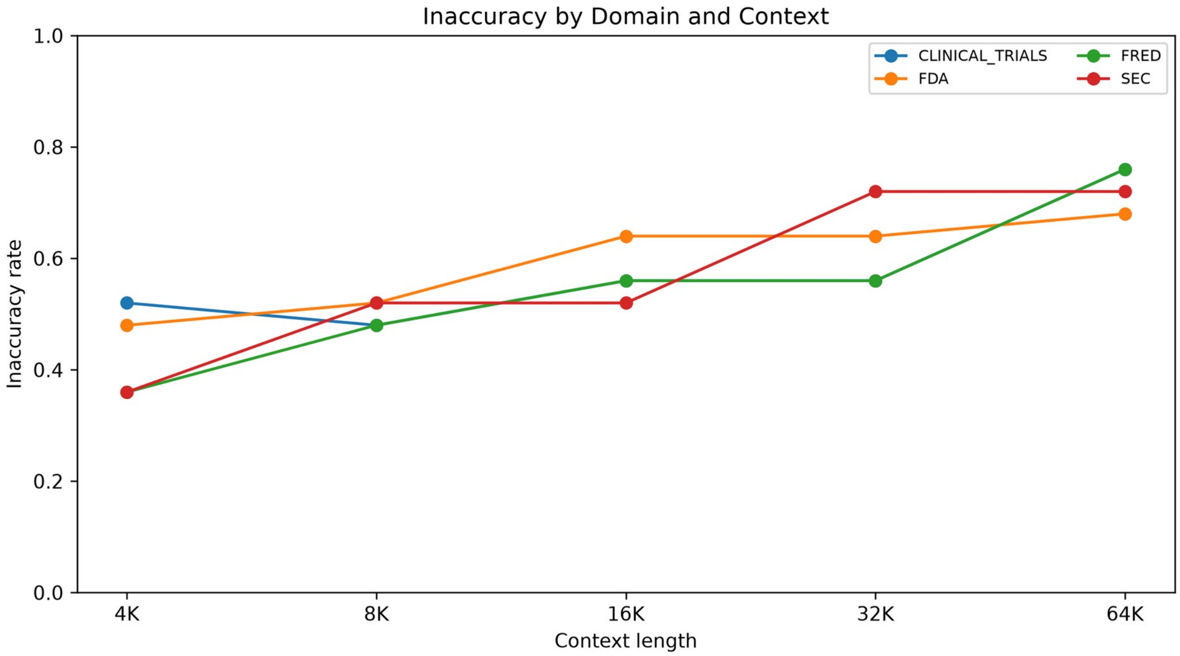
Group	Context	N	Correct	Inaccurate
DIRECT_RETRIEVAL	16K	20	70.0%	30.0%
DIRECT_RETRIEVAL	32K	20	65.0%	35.0%
DIRECT_RETRIEVAL	4K	20	95.0%	5.0%
DIRECT_RETRIEVAL	64K	20	70.0%	30.0%
DIRECT_RETRIEVAL	8K	20	80.0%	20.0%
ENTITY_UNIT_BINDING	16K	19	57.9%	42.1%
ENTITY_UNIT_BINDING	32K	19	57.9%	42.1%
ENTITY_UNIT_BINDING	4K	19	73.7%	26.3%

ENTITY_UNIT_BINDING	64K	19	31.6%	68.4%
ENTITY_UNIT_BINDING	8K	19	63.2%	36.8%
RETRIEVAL_CALCULATION	16K	30	6.7%	93.3%
RETRIEVAL_CALCULATION	32K	30	10.0%	90.0%
RETRIEVAL_CALCULATION	4K	30	13.3%	86.7%
RETRIEVAL_CALCULATION	64K	30	10.0%	90.0%
RETRIEVAL_CALCULATION	8K	30	6.7%	93.3%
TEMPORAL_VERSION	16K	11	54.5%	45.5%
TEMPORAL_VERSION	32K	11	45.5%	54.5%
TEMPORAL_VERSION	4K	11	90.9%	9.1%
TEMPORAL_VERSION	64K	11	27.3%	72.7%
TEMPORAL_VERSION	8K	11	90.9%	9.1%
UNANSWERABLE	16K	20	50.0%	50.0%
UNANSWERABLE	32K	20	30.0%	70.0%
UNANSWERABLE	4K	20	50.0%	50.0%
UNANSWERABLE	64K	20	5.0%	95.0%
UNANSWERABLE	8K	20	50.0%	50.0%

11. Domain Results

Domain analyses are exploratory because each domain contributes 25 families.

Inaccuracy by Domain & Context.



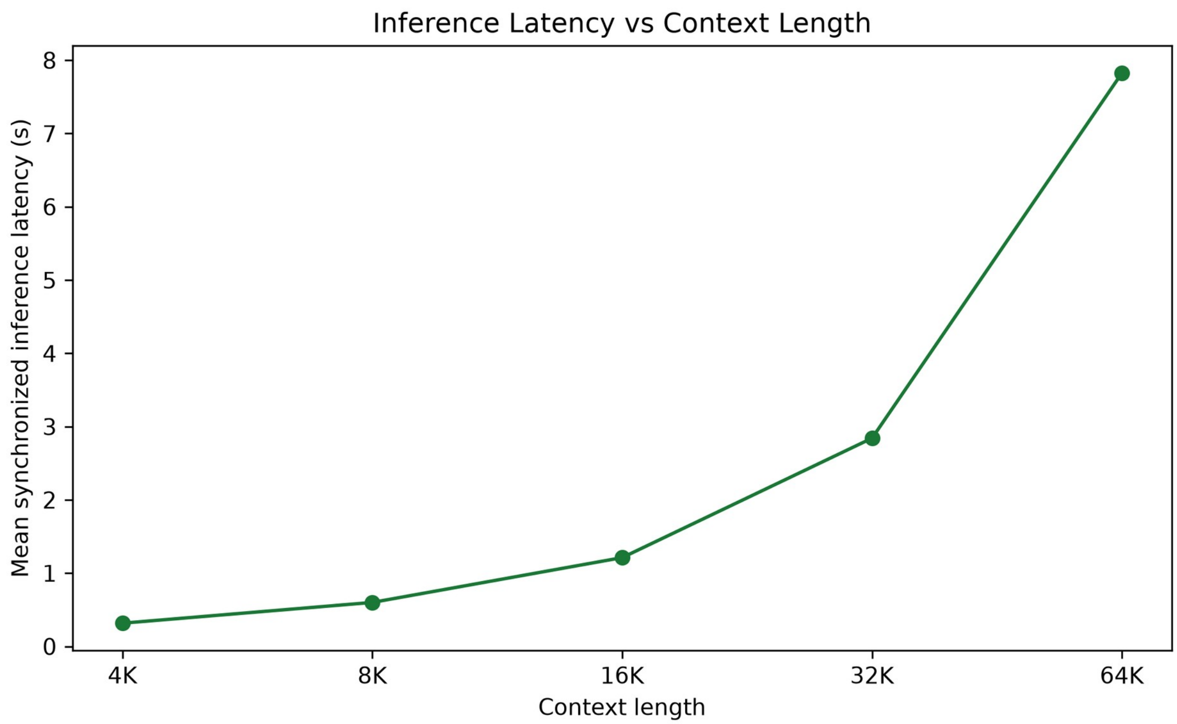
Subgroup results are exploratory because cells are smaller than the primary repeated-measures analysis.

Table 7. Inaccuracy by domain and context.

Group	Context	N	Correct	Inaccurate
CLINICAL_TRIALS	16K	25	44.0%	56.0%
CLINICAL_TRIALS	32K	25	44.0%	56.0%
CLINICAL_TRIALS	4K	25	48.0%	52.0%
CLINICAL_TRIALS	64K	25	24.0%	76.0%
CLINICAL_TRIALS	8K	25	52.0%	48.0%
FDA	16K	25	36.0%	64.0%
FDA	32K	25	36.0%	64.0%
FDA	4K	25	52.0%	48.0%
FDA	64K	25	32.0%	68.0%
FDA	8K	25	48.0%	52.0%
FRED	16K	25	44.0%	56.0%
FRED	32K	25	44.0%	56.0%
FRED	4K	25	64.0%	36.0%
FRED	64K	25	24.0%	76.0%
FRED	8K	25	52.0%	48.0%
SEC	16K	25	48.0%	52.0%
SEC	32K	25	28.0%	72.0%
SEC	4K	25	64.0%	36.0%
SEC	64K	25	28.0%	72.0%
SEC	8K	25	48.0%	52.0%

12. Inference-Time Analysis

Figure 3. Inference latency vs context length.



Mean synchronized generation latency rises as input length increases.

Table 8. Inference latency summary.

Context	Mean latency (s)	Median latency (s)	P95 latency (s)	Total latency (s)	Mean input tokens	Mean output tokens
4K	0.31732395038008 68	0.3150759311392903	0.34984864559955 89	31.73239503800869	4265.73	8.47
8K	0.60078134991228 58	0.5982510871253908	0.63596663670614 36	60.07813499122858	8329.19	8.42
16K	1.21216063832864 16	1.2097398359328508	1.26001920602284 37	121.21606383286417	16440.02	8.31
32K	2.84237699097953 73	2.8350786096416414	2.93536414308473 46	284.23769909795374	32671.02	8.37
64K	7.823902476206422	7.810557162389159	7.952514025289565	782.3902476206422	65120.48	8.3

Latency grows sharply with context length. This analysis is descriptive and should not be interpreted as causal evidence about reliability.

13. Interpretation, Limitations, and Conclusion

Longer contexts substantially reduce factual reliability in Llama 3.2 3B under this benchmark. Overall inaccuracy increased from 43% at 4K to 73% at 64K, and the odds of an inaccurate answer increased by approximately 35% per context doubling.

- The experiment includes 100 question families and 500 observations.
- Only one model was evaluated: Llama 3.2 3B Instruct.
- Experiment C covers contexts through 64K.
- The benchmark uses four factual/structured data domains and intentionally strong same-domain distractors.
- Experiment C uses an answer-only format and therefore has no evidence-selection metric.
- Grading is deterministic with one documented manual adjudication.
- Subgroup analyses have smaller samples and are exploratory.
- GEE was used instead of mixed-effects logistic regression for reliable repeated-measures fitting.
- Results may not generalize to larger, proprietary, or differently prompted models.

14. Methodological Provenance

Table 9. Provenance.

Item	Value
Final scored dataset hash	6fdfaa035b5da2211e813353916902c871e783ecfa993615db672f62bcb8e327
Frozen grader hash	d9282a0ccc50daba3bfd232c058dfbab63a5c19a7323d585a7c2236b3a6c4ba8
Analysis directory	data/analysis_experiment_c_final_v1
Statistical method	GEE logistic regression clustered by question_family_id; paired McNemar tests
Bootstrap CIs	Family-clustered bootstrap; seed 20260810; 5000 replicates
No reruns	Inference and grading were not rerun for this report.

Appendix A. Detailed Tables

Appendix Table A1. Full question-type results.

Group	Context	N	Correct	Inaccurate
DIRECT_RETRIEVAL	16K	20	70.0%	30.0%
DIRECT_RETRIEVAL	32K	20	65.0%	35.0%
DIRECT_RETRIEVAL	4K	20	95.0%	5.0%
DIRECT_RETRIEVAL	64K	20	70.0%	30.0%
DIRECT_RETRIEVAL	8K	20	80.0%	20.0%
ENTITY_UNIT_BINDING	16K	19	57.9%	42.1%
ENTITY_UNIT_BINDING	32K	19	57.9%	42.1%
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RETRIEVAL_CALCULATION	8K	30	6.7%	93.3%
TEMPORAL_VERSION	16K	11	54.5%	45.5%
TEMPORAL_VERSION	32K	11	45.5%	54.5%
TEMPORAL_VERSION	4K	11	90.9%	9.1%
TEMPORAL_VERSION	64K	11	27.3%	72.7%
TEMPORAL_VERSION	8K	11	90.9%	9.1%
UNANSWERABLE	16K	20	50.0%	50.0%
UNANSWERABLE	32K	20	30.0%	70.0%
UNANSWERABLE	4K	20	50.0%	50.0%
UNANSWERABLE	64K	20	5.0%	95.0%

UNANSWERABLE	8K	20	50.0%	50.0%

Appendix Table A2. Full domain results.

Group	Context	N	Correct	Inaccurate
CLINICAL_TRIALS	16K	25	44.0%	56.0%
CLINICAL_TRIALS	32K	25	44.0%	56.0%
CLINICAL_TRIALS	4K	25	48.0%	52.0%
CLINICAL_TRIALS	64K	25	24.0%	76.0%
CLINICAL_TRIALS	8K	25	52.0%	48.0%
FDA	16K	25	36.0%	64.0%
FDA	32K	25	36.0%	64.0%
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FRED	32K	25	44.0%	56.0%
FRED	4K	25	64.0%	36.0%
FRED	64K	25	24.0%	76.0%
FRED	8K	25	52.0%	48.0%
SEC	16K	25	48.0%	52.0%
SEC	32K	25	28.0%	72.0%
SEC	4K	25	64.0%	36.0%
SEC	64K	25	28.0%	72.0%
SEC	8K	25	48.0%	52.0%